How sharp is our soil-moisture map, really?

A plain-language guide to the effective-resolution test (runbook §70)

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0. The question in one paragraph

Our model draws a soil-moisture map on a grid of 20 m × 20 m pixels. That does *not* mean the map really knows what happens every 20 m. A photo can have millions of pixels and still be blurry. The **effective resolution** is the size of the smallest detail that is (1) actually present in the map and (2) actually correct. This document explains how we measured both, and what we found.

In one sentence: The map has 20 m pixels, but we want to know the size of the smallest detail that is real.

1 What the model produces

- The world is cut into square **tiles** of 2.24 km × 2.24 km.
- For each tile and each day, the model outputs a map of 112 × 112 pixels (112 × 20 m = 2.24 km). It does this for three depths: 0–10, 10–30 and 30–100 cm.
- Inside the model, each tile is first described by a coarser grid of 14 × 14 **tokens**. Each token covers 160 m × 160 m. These tokens carry most of the satellite information. A separate "fine path" also feeds 20 m satellite pixels into the last stage.
- We test on **TxSON**, a dense sensor network in Texas: 40 soil-moisture stations, many of them only a few hundred metres apart. 30 of the tiles were never used for training ("held-out"). All numbers below come from those 30.

2 Removing the pattern that never changes

Figure 1 shows one real tile. Panel (a) is the *average* map over all days. It has a clear pattern, but earlier work (§56) showed that this pattern is almost the same every day: the model paints it regardless of weather. A fixed pattern tells us nothing about whether the model follows real wetting and drying.

So, for every pixel, we subtract that pixel's own long-term average. What is left is the **anomaly**: "today this spot is wetter (blue) or drier (red) than it usually is" (panel c). *Every measurement in this document is made on anomaly maps.*

In one sentence: We throw away the part of the map that never changes and study only the day-to-day changes.

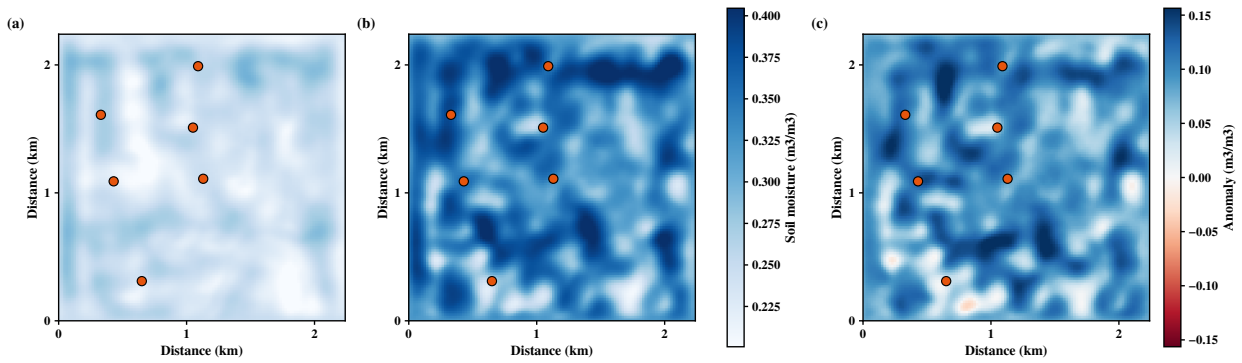


Figure 1: One held-out tile (TxSON CR200-18, 0–10 cm). (a) The average map over all days, which is the fixed pattern. (b) The map on one day (the day with the most spatial contrast). (c) That day minus the average: the anomaly. Orange dots are the real sensor stations inside this tile.

3 Three ways to ask "how fine is the detail?"

All three tools below measure the same property from different angles.

3.1 The variogram: "how different are two points that are h metres apart?"

Take many pairs of points at the same distance h (say 100 m). For each pair, measure how much their values differ, then average. Repeat for h = 20 m, 40 m, …, 1.6 km. The resulting curve is the **variogram**. We divide it by the total variance, so 0 means "identical" and about 1 means "completely unrelated".

Figure 2 shows two made-up maps to build intuition:

- **Smooth field (a):** neighbours look alike. The curve starts at 0 and rises slowly. It reaches its plateau only at several hundred metres. The distance where it flattens is called the **range**: beyond it, points are as different as two random points.
- **Rough field (b):** even neighbouring pixels differ, so the curve *jumps up immediately*. The jump at very short distance is called the **nugget**. It means there is detail finer than we can see, or simply noise.

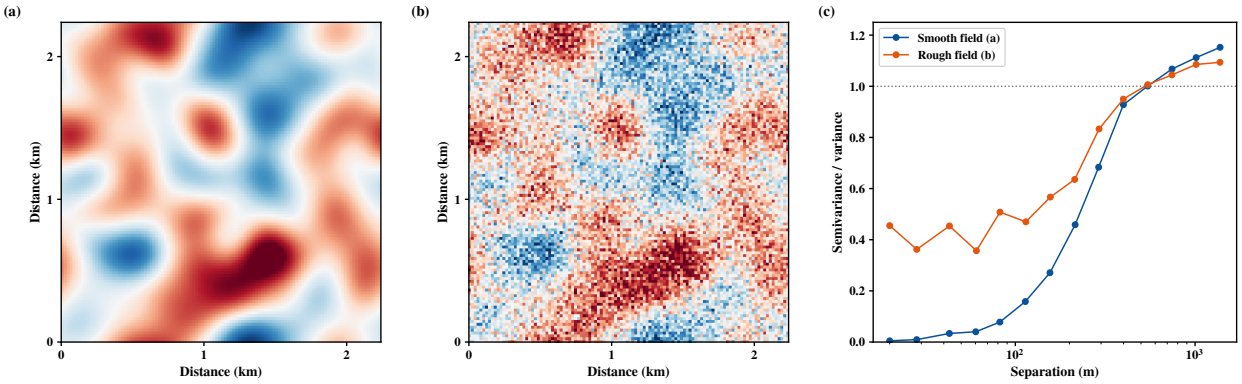


Figure 2: Toy example, not model output. (a) A smooth field with blobs a few hundred metres wide. (b) The same field with 20 m-scale noise added. (c) Their variograms: the smooth field rises gradually and has no nugget; the rough field jumps up at the first lag.

In one sentence: Where the variogram flattens (the range) is the size of the blobs in the map.

3.2 The spectrum: "how much of the map is made of waves of each size?"

Any map can be written as a sum of waves of different lengths: long gentle waves (2 km) plus shorter ripples (40 m). The **spectrum** measures how strong each wave length is. A smooth map is strong in long waves and has almost nothing in short ones. The spectrum and the variogram carry the same information; one speaks in wave lengths, the other in distances.

3.3 The blur test: "what do I lose if I blur the map?"

Average the map over blocks of $b \times b$ metres and measure how much of its variation survives. If blurring at 80 m keeps 100 %, the map had no detail at 80 m. The block size where the variation starts to disappear is where the detail lives.

4 Why 160 m and 320 m are marked on the figure

The model's tokens form a 160 m grid. Figure 3 shows what such a grid can and cannot hold.

- To draw a wave you need at least one wet cell and one dry cell, so at least **two cells per wave**. With 160 m cells, the shortest wave the grid can hold is $2 \times 160 = \mathbf{320\,m}$ (panel a). This limit is called the **Nyquist** limit.
- A wave shorter than that, for example 100 m (panel b), falls *inside* the cells. Each cell only keeps its average, which is close to zero, so the wave is lost.

So if the map's detail comes from the tokens, we *expect* it to stop at around 320 m wave length (160 m in distance). If the 20 m fine path added real detail, we would see power at shorter waves too.

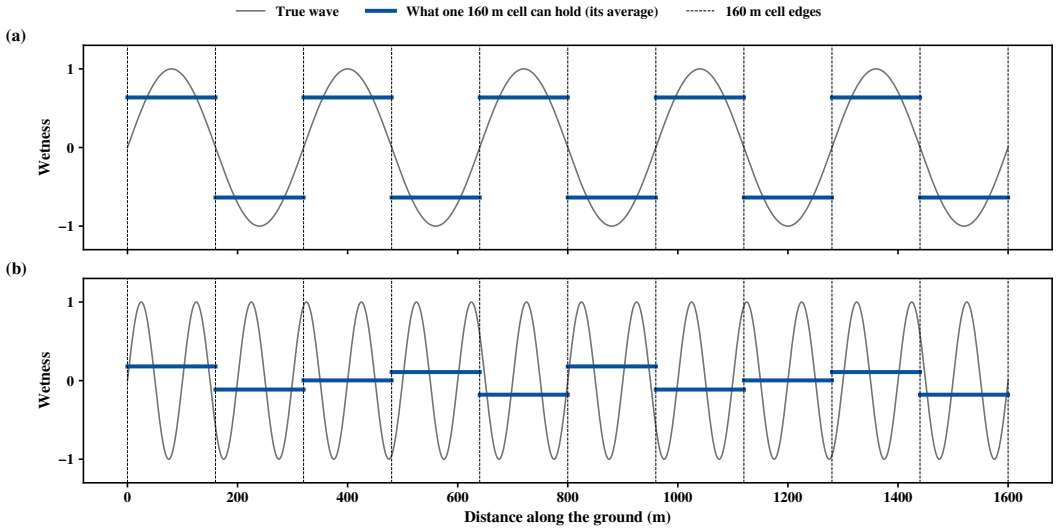


Figure 3: Toy example. (a) A 320 m wave on a 160 m grid: the cell averages (blue) still go up and down, so the grid can carry this wave. (b) A 100 m wave: the cell averages are almost flat, so the grid cannot carry it.

In one sentence: A 160 m grid can only draw waves 320 m long or longer; that is why those two lines are on the plots.

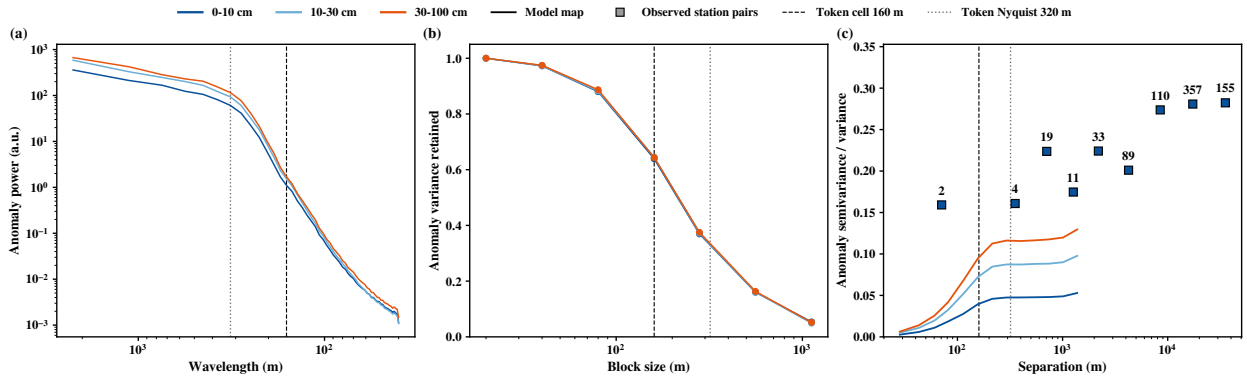


Figure 4: Result over the 30 held-out TxSON tiles. Colours are depths (dark blue 0–10 cm, light blue 10–30 cm, orange 30–100 cm). Dashed line: token cell, 160 m. Dotted line: token Nyquist, 320 m. (a) Spectrum of the anomaly maps. (b) Variation retained after blurring. (c) Variogram: lines are the model map; squares are real station pairs at 0–10 cm, with the number of pairs written above.

5 The result figure, panel by panel

Panel (a), spectrum. Read from left (long waves) to right (short waves). The curve is high and slowly falling down to about 320 m (dotted line), then it **drops about 100 times** by 160 m (dashed line). Below that there is almost nothing. *Meaning:* the map's day-to-day pattern is made of blobs about 300 m and larger, which is what the 160 m token grid can draw. The 20 m fine path adds almost nothing.

Panel (b), blur test. Blurring the map into 40 m blocks keeps 97 % of its variation; 80 m keeps 88 %, 160 m 64 %, 280 m 37 % and 560 m 16 %. *Meaning:* there is almost no detail at 20–80 m; the detail lives between about 160 and 500 m.

Panel (c), variogram.

- **Lines (model):** start near 0 (neighbouring pixels are near-copies), rise until about 250 m, then go flat. So the **range of the map is about 250 m**.
- **Height of the flat part:** 0.05 at 0–10 cm. Within a tile, all points move almost together: only about 5 % of their daily change differs from point to point.
- **Squares (real stations):** 0.16–0.22 at distances below 3 km and about 0.28 at 6–50 km. Real points disagree about **4 times more** than the model's points do. Part of the real difference is sensor noise, so 4× is an upper limit on the gap.

A note on the dotted 320 m line: it is exact only in panel (a), where the axis is a wave length. In panels (b) and (c) the axis is a block size or a distance, so the matching marker there is the dashed 160 m line.

In one sentence: The map contains detail down to about 250 m, and it is much smoother than the real ground.

6 Present is not the same as correct

The figure shows what detail the map *has*. It cannot show whether that detail is *right*. Three extra tests, compared against the real stations, answer that.

In one sentence: Blurring the map to the whole tile does not hurt it, so the detail inside a tile is not skill.

Test	What we did	What happened
C: blur the output	Blur the predicted map around each station, from 20 m up to the whole tile (2.24 km), and recompute the error against the station.	The error **never gets worse**, even with the whole tile blurred to one value (0.0452 $\rightarrow$ 0.0443 m^3/m^3). The fine detail adds no accuracy.
Control	Run the same test on a made-up "perfect" map that truly has 160 m detail.	The test answers 280 m. So the test *can* detect fine resolution when it exists.
B: blur the input	Blur the satellite inputs before they enter the model (40 m to 2.24 km) and predict again.	The map **changes a lot** (30 % of its spatial variation already at 40 m, 58 % at 160 m), but the station error **does not change**. The fine inputs reshape the picture without making it more accurate.
A: station pairs	For pairs of stations, compare the predicted difference with the observed difference.	Long-term differences: correlation about 0. Day-to-day differences: a weak but real signal at 500–1000 m (correlation 0.08, 95 % range 0.02–0.11).

Table 1: The correctness tests (0–10 cm only; deeper layers have too few station pairs).

7 Conclusion

1. The map **contains** detail down to about **250 m**. That is set by the model's 160 m token grid, not by the 20 m output pixels.
2. The detail is **not correct**. Removing it entirely does not make the predictions worse. The **effective resolution is about one tile, 2 km**, at 0–10 cm.
3. Within a tile the model draws **texture**: the inputs move it around, but it does not follow the real differences between nearby points. It is also about 4 times smoother than the real ground.

Glossary

Anomaly A pixel's value today minus its own long-term average.

Effective resolution The size of the smallest detail that is both present and correct.

Nugget The jump of a variogram at very short distance: detail finer than the grid, or noise.

Nyquist limit The shortest wave a grid can carry: two grid cells.

Range The distance where a variogram flattens; roughly the size of the blobs in the map.

Spectrum How strong each wave length is in a map.

Token One 160 m $\times$ 160 m cell of the model's internal 14×14 description of a tile.

ubRMSE The typical prediction error after removing the constant offset between model and station.

Variogram Half the average squared difference between two points, as a function of their distance.